

BORZYA, Russian FederationWMO: **309650**Lat: **50.400N**Lon: **116.517E**Elev: **676**StdP: **93.47**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.4	-37.0	-41.5	0.1	-39.0	-39.2	0.1	-36.8	9.0	-16.1	7.9	-17.0	1.0	70	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	31.2	17.6	29.2	17.0	27.1	16.5	19.7	26.8	18.8	25.3	18.0	24.0	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.7	21.7	16.8	13.0	21.0	15.9	12.3	20.4	59.5	27.0	56.6	25.3	53.8	24.0	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.0	8.5	7.4	-41.6	34.2	3.0	2.9	-43.8	36.2	-45.6	37.9	-47.3	39.5	-49.5	41.6	
(5)				-41.6	21.4	3.0	1.2	-43.7	22.3	-45.5	23.0	-47.1	23.6	-49.3	24.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.3	-25.5	-20.6	-10.0	1.7	9.8	17.0	19.6	17.0	9.8	0.0	-13.0	-22.8
	DBStd	16.59	5.14	5.85	6.81	5.00	4.57	3.88	2.97	3.35	4.42	5.13	6.59	5.28
	HDD10.0	4963	1099	858	621	252	58	2	0	1	56	311	689	1017
	HDD18.3	7282	1357	1091	879	499	267	67	18	64	258	568	939	1275
	CDD10.0	839	0	0	0	3	53	213	299	218	50	2	0	0
	CDD18.3	116	0	0	0	0	3	29	58	23	2	0	0	0
	CDH23.3	1646	0	0	0	4	82	491	696	327	46	0	0	0
	CDH26.7	535	0	0	0	1	21	172	238	95	8	0	0	0
Wind	WSAvg	2.8	1.7	2.1	3.0	4.0	4.0	3.1	2.8	2.7	3.0	3.0	2.3	1.8
Precipitation	PrecAvg	296	2	2	3	8	18	47	92	77	38	9	4	4
	PrecMax	432	5	14	15	29	70	126	182	193	118	47	12	11
	PrecMin	157	0	0	0	0	1	6	3	24	0	0	0	0
	PrecStd	70	1	3	3	7	14	30	41	35	28	9	3	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.4	-0.4	11.3	22.8	29.5	34.6	34.5	32.1	27.6	19.2	6.1	-5.9
		MCWB	-9.5	-3.5	3.9	9.6	14.6	17.3	18.4	18.4	14.6	9.1	1.3	-7.3
	2%	DB	-12.0	-4.5	7.0	18.0	25.5	30.7	31.5	29.4	24.3	14.8	2.2	-10.1
		MCWB	-12.8	-6.5	1.1	7.5	12.6	16.3	18.3	17.4	13.5	6.5	-1.4	-11.0
	5%	DB	-14.5	-7.4	4.1	14.6	22.6	28.4	29.4	27.0	21.4	11.7	-0.4	-12.4
		MCWB	-15.2	-8.8	-0.6	5.8	11.2	15.7	18.0	17.1	12.3	5.0	-2.9	-13.1
	10%	DB	-16.8	-10.3	1.2	11.7	19.8	26.0	27.1	24.6	18.9	8.9	-3.1	-14.4
		MCWB	-17.3	-11.3	-2.4	4.2	9.8	15.3	17.5	16.4	11.2	3.6	-5.0	-15.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.4	-3.6	4.1	10.2	16.0	19.3	21.2	20.5	16.4	9.3	1.6	-7.4
		MCDB	-8.5	-0.9	10.8	21.6	24.3	29.2	27.5	28.0	23.4	18.7	5.8	-6.0
	2%	WB	-12.8	-6.5	1.4	8.0	13.6	18.1	20.1	19.1	14.9	7.0	-0.9	-11.0
		MCDB	-12.0	-4.6	6.5	16.7	23.3	26.0	27.2	24.9	20.8	13.4	1.9	-10.1
	5%	WB	-15.2	-8.8	-0.6	6.2	11.9	17.2	19.3	18.2	13.6	5.4	-2.9	-13.1
		MCDB	-14.5	-7.4	3.7	13.9	21.5	24.9	26.1	23.8	19.8	11.1	-0.5	-12.4
	10%	WB	-17.3	-11.2	-2.2	4.4	10.5	16.3	18.6	17.3	12.0	3.9	-5.0	-15.0
		MCDB	-16.8	-10.3	1.2	11.2	18.8	23.8	24.8	22.7	18.0	8.7	-3.2	-14.4
Mean Daily Temperature Range	5% DB	MDBR	13.5	15.6	15.0	14.0	14.4	13.6	12.1	12.5	13.8	12.7	12.5	12.8
		MCDBR	14.8	17.1	17.1	19.3	19.2	18.1	16.5	17.0	18.3	17.4	13.9	13.7
		MCWBR	13.9	15.2	12.0	11.1	9.3	7.1	6.0	7.0	8.9	10.5	10.7	12.9
	5% WB	MCDBR	14.6	17.0	16.5	18.4	17.1	14.0	12.3	12.1	14.4	15.8	13.8	13.7
		MCWBR	13.8	15.2	11.8	10.8	8.7	6.1	5.4	5.8	7.3	9.7	10.8	12.9
Clear-Sky Solar Irradiance	taub	0.253	0.279	0.320	0.410	0.484	0.467	0.421	0.388	0.343	0.311	0.281	0.260	
	taud	2.226	2.172	2.154	2.040	1.883	2.050	2.258	2.324	2.389	2.379	2.274	2.188	
	Ebn at Noon	787	852	876	826	776	796	831	843	849	811	738	705	
	Edh at Noon	70	97	121	153	188	160	128	114	96	80	67	63	
All-Sky Solar Radiation	RadAvg	1.61	2.75	4.27	5.32	6.06	6.24	5.79	5.18	4.10	2.83	1.65	1.20	
	RadStd	0.07	0.13	0.18	0.30	0.41	0.55	0.37	0.36	0.28	0.14	0.08	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)		N/A	N/A	-2.20	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	-2.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page